

Hybrid Adaptive Quantum Particle Swarm Optimization for High-Resolution 2-D DOA Estimation Under Adverse Conditions

FAWAD ZAMAN¹ (Senior Member, IEEE), SADIQ AKBAR², AMIR RASHID CH³,
SHAFQAT ULLAH KHAN⁴, AND GRIDSADA PHANOMCHOENG^{1,5} (Senior Member, IEEE)

¹Department of Mechanical Engineering, Faculty of Engineering, Chulalongkorn University, Bangkok 10330, Thailand

²Department of Electronics, University of Peshawar, Peshawar, KP 25120, Pakistan

³Department of Electrical Engineering, COMSATS University, Islamabad 44000, Pakistan

⁴Department of Electronics, University of Buner, Swari 44000, Pakistan

⁵Center of Excellence for Human-Robot Collaboration and Systems Integration, Chulalongkorn University, Bangkok 10330, Thailand

CORRESPONDING AUTHOR: G. PHANOMCHOENG (gridsada.phanomchoeng@gmail.com)

This work was supported in part by the Second Century Fund (C2F), Chulalongkorn University, Bangkok, Thailand; in part by the 111th Anniversary Engineering Research Catalyst Fund Toward U Top 100; and in part by the Immersive Technology Lab (IMT Lab), Faculty of Engineering, Chulalongkorn University.

ABSTRACT Precise joint estimation of elevation-azimuth angles in planar phased-array radar is one of the hot research areas that become challenging particularly for closely spaced targets, low signal-to-noise ratio (SNR), and limited snapshots availability conditions. In such adverse situation, the performance of the traditional subspace-based techniques such as Two Dimensional Multiple Signal Classification (2D-MUSIC) significantly degraded due to grid discretization limitations and noise subspace leakage, while classical Particle Swarm Optimization (PSO) algorithm suffers due to premature convergence and insufficient exploitation ability. To overcome these issues, this work proposes a novel methodology called Hybrid Adaptive Quantum Particle Swarm Optimization (HAQPSO) for high-resolution two-dimensional (2D) direction-of-arrival (DOA) estimation under the said adverse conditions. The proposed HAQPSO implements the 2D DOA estimation as a multidimensional nonlinear optimization problem based on a determinant-normalized subspace orthogonality cost function, increasing discrimination capability for closely spaced targets. To balance global and local exploration and exploitation, respectively, an adaptive contraction–expansion coefficient is incorporated into the quantum-behaved swarm update. A local deterministic coordinate descent refinement stage is hybridized with the global quantum search, which further improves convergence precision and mitigates stagnation. Contrary to the traditional PSO, the proposed technique utilizes a mean based driven logarithmic position update mechanism that enhances population diversity and robustness in noisy situation with no requirements of exhaustive spectral grid scanning as in 2D-MUSIC. Comprehensive Monte Carlo simulations are conducted on a uniform rectangular array to demonstrate the performance of proposed HAQPSO that consistently gains lower Root-Mean-Square Error (RMSE) and higher estimation accuracy compared to standard PSO and other state of the art algorithms across a wide SNR range. In particular, for closely spaced targets with 3 degree angular separation at SNR levels as low as -5 dB, the proposed algorithm obtains higher estimation accuracy and improved target resolvability.

INDEX TERMS Antenna array, direction of arrival, phased array radar, planar array, quantum particle swarm optimization, heuristic computation.

I. INTRODUCTION

DIRECTION of Arrival (DOA) estimation is a key problem in array signal processing, with vital applications in electronic surveillance, wireless communications, radar,

and sonar systems [1], [2], [3], [4]. Efficient estimation of DOA assists the beamformer network in minimizing jammer interference, while positioning the main beam in the desired look direction, which is critical for both defense and civilian

radar systems [5], [6], [7], [8]. DOA estimation using one dimensional sensor arrays is relatively easy, where the array manifold vectors (AMV) depend only on the elevation angle whose range is limited to 180 degrees [9], [10], [11]. In order to cover the entire 360 degrees, multi-dimensional sensor arrays are to be used, where the AMV are a function of both elevation and azimuth angles [12], [13], [14], [15]. Among various multi-dimensional array geometries, planar phased arrays received special attention among the research communities because they offer two-dimensional (2D) angular coverage, accepting real-time estimation of azimuth and elevation angles [16], [17], [18], [19].

Regardless of extensive applications, estimation of 2D DOA remains a challenging problem, specifically under adverse operating conditions such as low signal-to-noise ratio (SNR), limited snapshot availability, closely located targets, and the pair matching issue between the corresponding angles for a specific target [20], [21], [22], [23], [24]. Traditional subspace-based techniques, such as multiple signal classification (MUSIC) and estimation of signal parameters via rotational invariance techniques (ESPRIT), are widely implemented for 2D DOA estimation. These methods exploit the orthogonality between signal and noise subspaces to achieve super resolution beyond the classical Rayleigh limit [25], [26], [27]. However, their applicability is under question especially in the presence of low SNR and generally requires a large number of snapshots to achieve the required angles. In addition, MUSIC requires exhaustive spectral scanning over the azimuth–elevation grid, which increases the computational burden and susceptibility to grid discretization errors [28], [29], [30], [31], [32]. Similarly, subspace leakage under low-SNR conditions can severely degrade estimate accuracy, resulting in a higher root-mean-square error (RMSE) and a reduced probability of resolution, especially for closely located targets separated by a few degrees [20], [33], [34], [35].

To address the said issues, evolutionary computing methods are proposed for joint 2D DOA estimation [36], [37], [38], [39]. These algorithms exploit the DOA estimation problem as a multidimensional nonlinear optimization problem, where the cost function calculates the difference between the desired output of the array and the estimated response of the system. These techniques not only have the ability to avoid local minima and grid scanning during the search space, but are also handy to tackle arbitrary array geometries [40]. However, these techniques, especially Particle Swarm Optimization (PSO), often face the problem of premature convergence and insufficient exploitation capability under complex adverse conditions, particularly when working in the presence of low SNR or when the targets are close to each other [41], [42]. Recent developments in quantum-inspired meta-heuristics have achieved improved global search skills and enhanced convergence behavior [43], [44], [45], [46]. Quantum-behaved PSO (QPSO) combines the principle of quantum mechanics with the particle update equations of PSO. It substitutes deterministic velocity updates

with probabilistic position evolution based on the particle's mean-best (mbest) position. It helps the algorithm bypass any local minima and converge to the desired global optima in the search space [47], [48], [49]. However, in the presence of multiple closely located targets or in the presence of low SNR, QPSO may still require fine-scale exploitation and hence demand the hybridization of a local refinement algorithm to handle these adverse conditions efficiently. In this context, this work proposes a novel hybrid methodology, in which a deterministic coordinate descent algorithm [50], [51] is combined with the global quantum search that further improves the results and mitigates stagnation. The proposed hybrid scheme implements the 2D DOA estimation as a multidimensional nonlinear optimization problem based on a determinant-normalized subspace orthogonality cost function, increasing discrimination capability for closely spaced targets. To balance global and local exploration and exploitation, respectively, an adaptive contraction–expansion coefficient is incorporated into the quantum-behaved swarm update. Contrary to traditional PSO, the proposed technique utilizes a mean based driven logarithmic position update mechanism that enhances population diversity and robustness in adverse environment without exhaustive spectral grid scanning as in 2D-MUSIC. Comprehensive Monte Carlo simulations are conducted on a uniform rectangular array to demonstrate the performance of proposed algorithm that consistently gains lower Root-Mean-Square Error (RMSE) and higher estimation accuracy compared to standard PSO and other state of the art techniques across a wide SNR range. In Particular, for closely spaced targets with 3 degree angular separation at SNR levels as low as -5 dB, the proposed algorithm obtains higher estimation accuracy and enhanced target resolvability.

The salient features of the current work are as follows:

- i. Development of an efficient and robust frame work for the joint 2D DOA estimation in Planar phased array radar system.
- ii. Design and Implementation of Hybrid Adaptive Quantum Particle Swarm Optimization (HAQPSO) for high-resolution 2D-DOA direction-of-arrival (DOA) estimation under adverse environments.
- iii. Design of a cost function based on the orthogonality of determinant-normalized subspace that measures the misalignment between the candidate steering vectors and the noise subspace.
- iv. Monte Carlo simulations are conducted to validate the usefulness of the proposed technique under conditions such as low SNR and closely spaced targets.

The remainder of the work is organized as follows: Section II describes the mathematical model for 2D DOA estimation of plane waves impinging on a rectangular sensor array. Section III is dedicated to the design and implementation of the proposed hybrid adaptive quantum particle swarm optimization algorithm. Section IV discusses the validity of the proposed scheme in terms of estimation accuracy, convergence analysis, root mean square error, and performance comparison with the classical PSO and other state of the

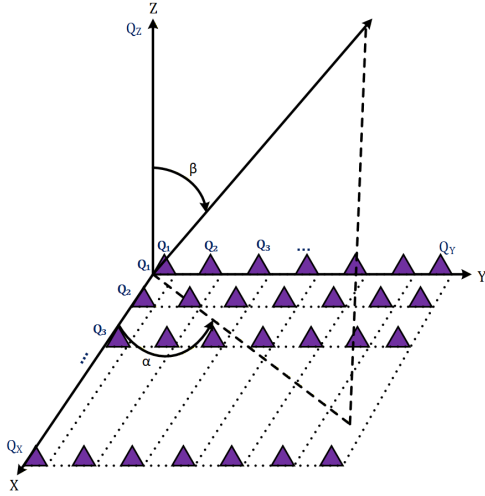


FIGURE 1. Planar array geometry.

art methods. Section V concludes the work by highlighting the key contributions and potential future directions of the proposed approach.

A. NOTATIONS

B, b	Scalar.
$\mathbf{b}$	Array manifold vector.
$\mathbf{B}$	Array Manifold Matrix.
$(\cdot)^*$	Complex conjugate.
$(\cdot)^H, (\cdot)^T$	Hermitian and transpose.
$\otimes$	Kronecker products.
$\mathbf{I}_Q$	Q -dimensional identity matrix.
β	Elevation angle.
α	Azimuth angle.
$\mathbb{E}\{\cdot\}$	Expectation operator.
$\text{diag}(\mathbf{B})$	Diagonal matrix with $\mathbf{B}$ as main diagonal.
$\text{Tr}(\mathbf{B})$	Trace of matrix $\mathbf{B}$.
$\lambda_{\min}(\mathbf{A})$	Minimum eigenvalue of matrix $\mathbf{B}$.
$\mathbb{R}, \mathbb{C}$	Set of real and complex numbers.

II. SYSTEM MODEL AND PROBLEM FORMULATION FOR PLANAR PHASED-ARRAY RADAR MODEL

In this section, we develop a mathematical model for plane waves impinging on a planar (rectangular) array, which is considered to be mounted on a passive phased array radar. The Planar Phased Array Radar (PPAR) plays a critical role in missile tracking, electronic surveillance, air traffic control, etc [14], [52], [53]. It has the ability to manage and track several targets at the same time and offers better resolution with rapid beam scanning capabilities. We have considered a uniform rectangular array (URA) having Q_x and Q_y sensors along the x -axis and y -axis, respectively, yielding a total of $Q = Q_x Q_y$ sensors as shown in fig.1. The uniform separation between two consecutive sensors in the array and the operating wavelength are respectively denoted by d and λ . Assume P narrowband independent far-field targets that arrived on the array having elevation and azimuth angles to

be β_p and α_p , respectively, where $p = 1, 2, \dots, P$. The signal received at the (q_x, q_y) -th sensor is given mathematically as:

$$z_{q_x, q_y}(t) = \sum_{p=1}^P s_p(t) b_{q_x, q_y}(\beta_p, \alpha_p) + n_{q_x, q_y}(t), \quad (1)$$

In equ(1), $s_p(t)$ is the amplitude of the p -th target, $b_{q_x, q_y}(\beta_p, \alpha_p)$ is the steering response of the sensor, and $n_{q_x, q_y}(t)$ is the additive white Gaussian noise, where $q_x = 0, 1, 2, \dots, Q_x - 1$ and $q_y = 0, 1, 2, \dots, Q_y - 1$. The Array Manifold Vector (AMV) belongs to the p -th target is given as:

$$\mathbf{b}_p = \mathbf{b}_y(\beta_p, \alpha_p) \otimes \mathbf{b}_x(\beta_p, \alpha_p), \in \mathbb{C}^{Q \times 1} \quad (2)$$

where, $\otimes$ represents Kronecker product. In equ(2),

$$\mathbf{b}_x(\beta_p, \alpha_p) = [1, e^{jKd\psi_p}, \dots, e^{jKd(Q_x-1)\psi_p}]^T \quad (3)$$

$$\mathbf{b}_y(\beta_p, \alpha_p) = [1, e^{jKd\zeta_p}, \dots, e^{jKd(Q_y-1)\zeta_p}]^T \quad (4)$$

where $\psi_p = \sin(\beta_p) \cos(\alpha_p)$ and $\zeta_p = \sin(\beta_p) \sin(\alpha_p)$. Similarly, for L snapshots, the output of all Q sensors is the $Q \times L$ data matrix given as:

$$\mathbf{Y} = \mathbf{B}\mathbf{S} + \mathbf{N}, \quad (5)$$

where, $\mathbf{B} = [\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_P] \in \mathbb{C}^{Q \times P}$ is the array manifold matrix, $\mathbf{S} \in \mathbb{C}^{P \times L}$ denotes the target amplitudes over L snapshots, $\mathbf{N} \in \mathbb{C}^{Q \times L}$ contains additive noise with zero-mean i.i.d. complex Gaussian entries of variance σ_n^2 .

Using equ(5), the sample covariance matrix can be given as:

$$\mathbf{R} = \frac{1}{L} \mathbf{Y} \mathbf{Y}^H = \mathbf{B} \mathbf{R}_s \mathbf{B}^H + \sigma_n^2 \mathbf{I}_Q, \quad (6)$$

where $\mathbf{R}_s = \frac{1}{L} \mathbb{E}[\mathbf{S} \mathbf{S}^H]$ is the target covariance matrix and $\mathbf{I}_Q$ is the $Q \times Q$ identity matrix. The goal is to estimate the elevation and azimuth angles i.e., $\beta = \{\beta_1, \beta_2, \dots, \beta_P\}$ and $\alpha = \{\alpha_1, \alpha_2, \dots, \alpha_P\}$ from the received data matrix $\mathbf{Y}$.

Subspace-based methods, such as MUSIC, use Eigen-Value Decomposition (EVD) of $\mathbf{R}$ to construct signal and noise subspaces:

$$\mathbf{R} = \mathbf{U}_s \mathbf{\Lambda}_s \mathbf{U}_s^H + \mathbf{U}_n \mathbf{\Lambda}_n \mathbf{U}_n^H, \quad (7)$$

where $\mathbf{U}_n$ and $\mathbf{U}_s$ correspond to the noise and signal subspaces, respectively. The spatial spectrum of MUSIC can be given as

$$P_{\text{MUSIC}}(\beta, \alpha) = \frac{1}{\mathbf{b}^H(\beta, \alpha) \mathbf{U}_n \mathbf{U}_n^H \mathbf{b}(\beta, \alpha)}. \quad (8)$$

However, it is computationally expensive due to large snapshots and exhaustive grid search requirements. In addition, it fails when the targets are correlated and also noise sensitive, especially under low-SNR conditions.

To overcome these issues, we reformulate DOA estimation as a nonlinear optimization problem:

$$\{\hat{\beta}, \hat{\alpha}\} = \arg \min_{\beta, \alpha} J(\beta, \alpha), \quad (9)$$

where the determinant-normalized subspace orthogonality cost function is introduced, which is discussed in the next section.

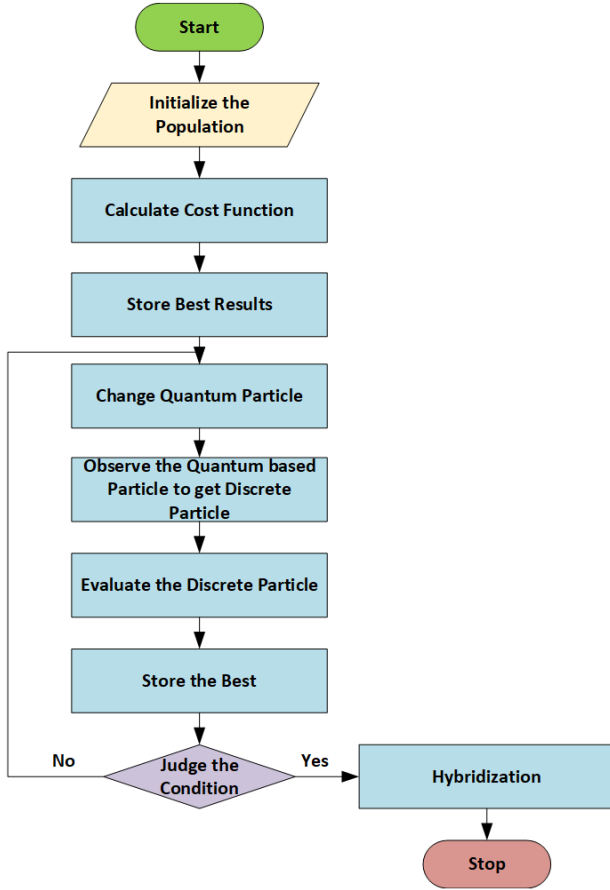


FIGURE 2. Flow chart of HAQPSO.

III. PROPOSED METHODOLOGY

Particle Swarm Optimization (PSO), inspired by bird flocking and fish schooling, is widely applied for nonlinear and multimodal optimization problems due to its simple structure and its strong global search ability [54]. However, the key issue related to traditional PSO is its premature convergence, particularly in high-dimensional search spaces [55], [56]. To address these shortcomings, the Quantum-behaved Particle Swarm Optimization (QPSO) algorithm is designed, where the principle of quantum mechanics is exploited to control particle movements in contrast to the traditional PSO that uses explicit velocity updates [57]. This concept improves the global search capability and provides a better compact parameterization compared to conventional PSO [58]. It also improves the balance of exploration and exploitation ability during the optimization process. The proposed Hybrid Adaptive Quantum Particle Swarm Optimization (HAQPSO) approach combines an adaptive quantum search strategy along with elitist preservation and deterministic local refinement, making it even appropriate for high resolution 2D DOA estimation particularly under low SNR and limited snapshot conditions. The flow chart of the proposed HAQPSO is shown in fig.2, its pseudo code is provided in appendix A, while its implementation steps for 2D DOA estimation are given below.

i. Initialization: The algorithm begins by generating a random population of M candidate solutions. Each particle corresponds to a potential set of P DOA pairs (β_p, α_p) . The m -th particle can be given as:

$$\mathbf{p}_m = [\beta_1, \alpha_1, \beta_2, \alpha_2, \dots, \beta_P, \alpha_P]_m, \quad m = 1, 2, \dots, M \quad (10)$$

where the lower and upper limits of the elevation angles are defined within the range $\beta_p \in [0, \pi/2]$ and the azimuth angles within $\alpha_p \in [0, 2\pi]$. Consequently, each particle is randomly initialized as:

$$\beta_p = \frac{\pi}{2} \cdot \text{rand}(), \quad (11)$$

$$\alpha_p = 2\pi \cdot \text{rand}(), \quad p = 1, 2, \dots, P. \quad (12)$$

ii. Cost Function: In order to estimate the 2D-DOA, the fitness (cost) function is designed that exploits the misalignment between candidate manifold vectors and the noise subspace:

$$J(\mathbf{p}_m) = \frac{\text{Tr}(\hat{\mathbf{B}}_m^H \mathbf{U}_n \mathbf{U}_n^H \hat{\mathbf{B}}_m)}{|\det(\hat{\mathbf{B}}_m^H \hat{\mathbf{B}}_m)| + \epsilon}, \quad (13)$$

where $\hat{\mathbf{B}}_m$ is the steering matrix corresponding to particle $\mathbf{p}_m$, $\mathbf{U}_n$ is the estimated noise subspace of the sample covariance, and ϵ is a small positive regularization constant. Minimizing $J(\mathbf{p}_m)$ ensures orthogonality to the noise subspace while resolving closely spaced sources. In equ(13), $\hat{\mathbf{B}} = [\hat{\mathbf{b}}_1, \hat{\mathbf{b}}_2, \dots, \hat{\mathbf{b}}_M] \in \mathbb{C}^{M \times P}$, where,

$$\hat{\mathbf{b}}_m = \hat{\mathbf{b}}_y(\hat{\beta}_m, \hat{\alpha}_m) \otimes \hat{\mathbf{b}}_x(\hat{\beta}_m, \hat{\alpha}_m), \quad (14)$$

and

$$\hat{\mathbf{b}}_x(\hat{\beta}_m, \hat{\alpha}_m) = [1, e^{j\pi\hat{\psi}_m}, \dots, e^{j\pi(M_x-1)\hat{\psi}_m}]^T \quad (15)$$

$$\hat{\mathbf{b}}_y(\hat{\beta}_m, \hat{\alpha}_m) = [1, e^{j\pi\hat{\zeta}_m}, \dots, e^{j\pi(M_y-1)\hat{\zeta}_m}]^T \quad (16)$$

where $\hat{\psi}_m = (\sin\hat{\beta}_m) \cos(\hat{\alpha}_m)$ and $\hat{\zeta}_m = \sin(\hat{\beta}_m) \sin(\hat{\alpha}_m)$.

iii. Adaptive Quantum Particle Swarm Update: In this step, quantum-behaved position update rule is applied to update the position of each particle. Let $\mathbf{i}_{\text{best}}$ denote the mean best position of the population:

$$\mathbf{i}_{\text{best}} = \frac{1}{M} \sum_{m=1}^M \mathbf{p}_m. \quad (17)$$

The particle position is updated as:

$$\mathbf{p}_m^{t+1} = \mathbf{i}_{\text{best}} + \text{sign}(\mathbf{r}) \cdot \gamma(t) \cdot |\mathbf{i}_{\text{best}} - \mathbf{p}_m^t| \cdot \ln\left(\frac{1}{\mathbf{u}}\right), \quad (18)$$

where $\mathbf{u} \sim \text{Uniform}(0, 1)$, $\mathbf{r} \sim \text{Bernoulli}(0.5)$, and $\gamma(t)$ is an adaptive contraction-expansion coefficient:

$$\gamma(t) = \gamma_0 \left(1 - \frac{t}{T}\right), \quad (19)$$

with γ_0 representing the initial quantum coefficient, while T is the maximum number of iterations. This update balances local exploitation and global exploration during the iterative search.

iv. Elitist Selection: Once the fitness of each particle is evaluated, the one with lower fitness is stored as an elite in this step as given as:

$$\mathbf{p}_{\text{elite}} = \arg \min_m J(\mathbf{p}_m). \quad (20)$$

In order to avoid deterioration of the best solution due to stochastic fluctuations, the elite particle is preserved across each iteration.

v. Hybridization: Although, the meta heuristic-based adaptive quantum particle swarm offers significant global exploration ability, yet it may have small residual estimation errors due to its stochastic search strategy. Therefore, to further enhance the estimation accuracy, a local search algorithm called deterministic coordinate descent (DCD) is hybridized with QPSO. In this hybridization process, a DCD algorithm is applied to the elite solution achieved by QPSO. For each angular dimension (β_p, α_p) , the particle is gradually perturbed with respect to a fixed step, and this move is accepted only if the fitness improves:

$$\beta_p \leftarrow \beta_p - \delta \cdot \text{sign}(\Delta J_\beta), \quad (21)$$

$$\alpha_p \leftarrow \alpha_p - \delta \cdot \text{sign}(\Delta J_\alpha), \quad (22)$$

where δ represents a small step size, while $\Delta J_\beta, \Delta J_\alpha$ denotes the change in the cost function along each angular dimension. This ensures better resolution estimation for targets that are closely spaced. The DCD is particularly appropriate for the proposed hybrid optimization framework for three main reasons. First, it offers a fine-grained local exploitation around the elite particle obtained during the global QPSO search. Secondly, it prevents the random oscillatory behavior that may stay in stochastic meta-heuristics near convergence. Third, it has minimal computational complexity compared to re-running a full population based optimization, since only one angular dimension is experienced at a time.

vi. Stopping Criterion: The iterative optimization process is stopped when the progress in the best fitness value between two consecutive iterations falls below a predefined threshold ε_s , or the maximum number of iterations T is completed. i.e.,

$$\left| J_{\text{best}}^{(t)} - J_{\text{best}}^{(t-1)} \right| < \varepsilon_s, \quad (23)$$

where $J_{\text{best}}^{(t)}$ denotes the minimum cost value obtained in the t -th iteration. This step avoids unnecessary computations once the swarm has converged significantly and improves the overall computational efficiency of the proposed HAQPSO algorithm.

vii. Storage: Store all the results obtained for later discussion, analysis, and comparison.

IV. SIMULATION RESULTS

In this section, several simulations are carried out to evaluate the effectiveness of the proposed HAQPSO framework. A uniform rectangular array (URA) is used with sensors $Q_x = 4$ and $Q_y = 4$ placed along the x - and y -axes, respectively, resulting in a total of $M = 16$ sensors. The spacing between two adjacent elements is set to $d = \lambda/2$,

TABLE 1. System specifications.

Sr.No.	Name	Specifications
1	Software	MATLAB 2022b
2	Operating System	WINDOWS 11 Pro, 64-bit, x64-based processor
3	Processor	11th Gen Intel(R) Core(TM) i7-1185G7 @ 3.00GHz (3.00 GHz)
4	RAM	16GB

where λ is the wavelength of the signals. Throughout the simulations, the maximum number of iterations is set to $T = 100$, the population size is maintained $N = 80$, while the temporal snapshots are limited to $L = 10$. Initially, the value of the contraction–expansion coefficient $\gamma_0 = 1$ and in order to obtain a balance between local exploitation and global exploration, its value decreases with iterations. The value of the regularization constant is set to be 10^{-3} which helps to prevent the numerical instability when the determinant term becomes very small as given in equ.(13). In addition, the proposed algorithm will terminate when the predefined value of MSE (10^{-6}) is obtained or the maximum number of iterations are completed. In the first part of the simulations, the resolution, convergence, and RMSE of HAQPSO is validated and the obtained results are compared with traditional PSO. In the second part of the simulations, the performance of HAQPSO is compared with 2D-MUSIC [20], Two dimensional Total Least Square Estimation of Signal Parameter Through Rotational Invariant Technique (2D TLS-ESPRIT) [14] and Orthogonal Matching Pursuit (OMP) [15] algorithm in the presence of adverse conditions. These adverse conditions include closely spaced targets, low SNR conditions, and limited snapshots availability. In the third and last part, a computational complexity analysis is carried out for HAQPSO and is compared with 2D-MUSIC [20], Two dimensional Total Least Square Estimation of Signal Parameter Through Rotational Invariant Technique (2D TLS-ESPRIT) [14] and Orthogonal Matching Pursuit (OMP) [15]. All the results obtained for elevation and azimuth angles are averaged over 100 independent trials. The simulations are performed on the machine whose specifications are provided in Table.1

Case a. Estimation Accuracy: In this case, the estimation accuracy of HAQPSO is evaluated for two, three, and four targets with SNR=5dB. Initially, two narrow band targets in the far field are considered whose true values of elevation and azimuth angles were $(\beta_1, \alpha_1) = (50^\circ, 110^\circ)$ and $(\beta_2, \alpha_2) = (70^\circ, 160^\circ)$. It can be seen from fig. 3, that the proposed HAQPSO achieved excellent estimation accuracy compared to traditional PSO.

In case of three targets, the true values of the angles were $(\beta_1, \alpha_1) = (40^\circ, 120^\circ)$, $(\beta_2, \alpha_2) = (65^\circ, 210^\circ)$ and $(\beta_3, \alpha_3) = (25^\circ, 300^\circ)$ and as shown in fig.4, the HAQPSO achieved once again very good accuracy compared to PSO. Similarly, fig.5, demonstrate the resolution of HAQPSO for four targets, where the true angle values were

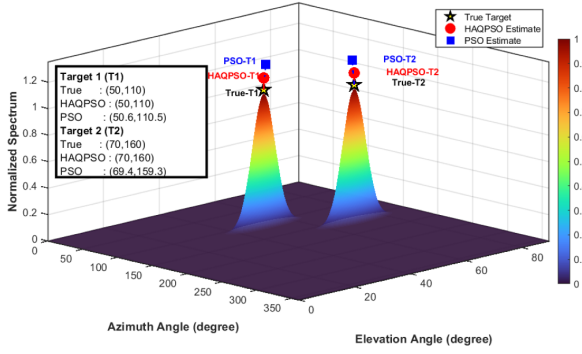


FIGURE 3. 2D DOA Estimation for two targets.

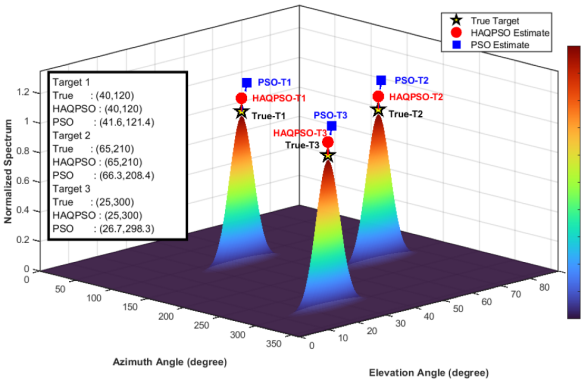


FIGURE 4. 2D DOA Estimation for three targets.

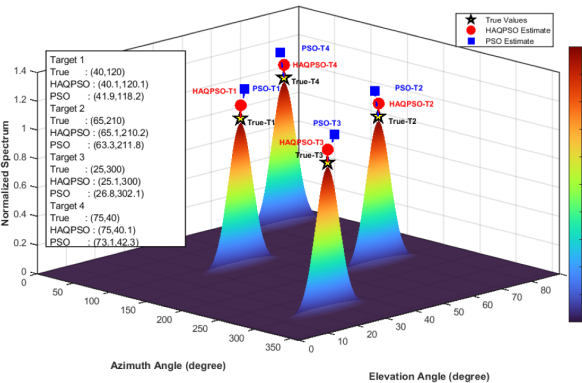


FIGURE 5. 2D DOA Estimation for four targets.

$(\beta_1, \alpha_1) = (40^\circ, 120^\circ)$, $(\beta_2, \alpha_2) = (65^\circ, 210^\circ)$, $(\beta_3, \alpha_3) = (25^\circ, 300^\circ)$ and $(\beta_4, \alpha_4) = (75^\circ, 40^\circ)$. As shown in fig.5, once again, HAQPSO achieved fairly good accuracy even for the increase in the number of unknowns, while the results of PSO degraded significantly in this case.

Case b. Convergence Analysis: In this case, the convergence of HAQPSO and PSO is discussed. From convergence, we mean how many times each algorithm achieved acceptable results out of the total number of trials for any particular case. Any trial is considered to be acceptable if the absolute estimation errors of both the elevation and azimuth angles do not exceed (1°) for all targets. The reported values belong to the number (or percentage) of successful trials out of the

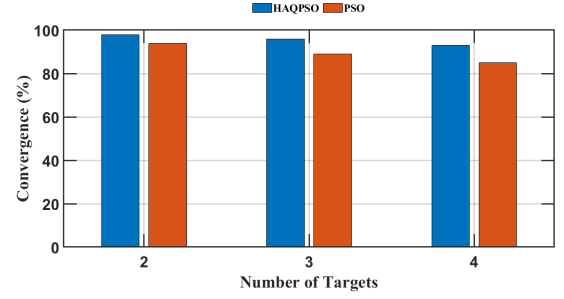


FIGURE 6. Convergence analysis.

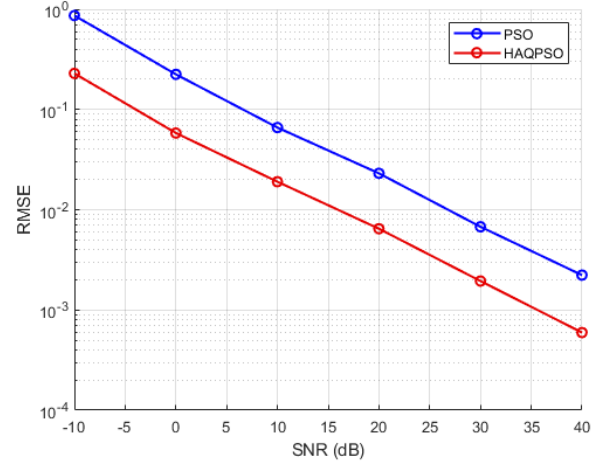


FIGURE 7. RMSE vs SNR.

the total 100 Monte Carlo runs. This analysis is important especially for stochastic based algorithms, where the outcome of the system can not be predicted accurately for the same inputs in each run. As shown in fig.6, the proposed hybrid framework, HAQPSO obtained better convergence for two, three, and four targets compared to PSO acting alone. The true values of angles and SNR were kept the same as in the previous case.

Case c. Root Mean Square (RMSE) Analysis: In this case, RMSE of HAQPSO is compared to PSO under different values of SNR. The following formula is used to evaluate the RMSE.

$$RMSE = \frac{1}{P} \sqrt{\sum_{p=1}^P [(\beta_p - \hat{\beta}_p)^2 + (\alpha_p - \hat{\alpha}_p)^2]}$$

For this case, two targets were considered whose true angles values were $(\beta_1, \alpha_1) = (20^\circ, 30^\circ)$, $(\beta_2, \alpha_2) = (23^\circ, 33^\circ)$. As shown in fig.7, HAQPSO has achieved better RMSE compared to PSO.

Case d. Proximity Effect: In this case, the proximity effect of HAQPSO is compared with those achieved by PSO and MUSIC [20] algorithms for three targets. In this case, three targets were considered that were closely spaced to each other. The true angles (DOAs) were $(\beta_1, \alpha_1) = (40^\circ, 70^\circ)$, $(\beta_2, \alpha_2) = (42^\circ, 72^\circ)$, and $(\beta_3, \alpha_3) = (44^\circ, 74^\circ)$, representing closely spaced targets in both elevation and azimuth domains. This experiment was conducted under low SNR condition

TABLE 2. Proximity effect analysis under low SNR for three targets.

Algorithm	T1 (β_1, α_1)	T2 (β_2, α_2)	T3 (β_3, α_3)
True DOAs	(40°, 70°)	(42°, 72°)	(44°, 74°)
MUSIC	(41.8°, 67.3°)	(44.4°, 74.8°)	(46.9°, 76.4°)
PSO	(37.1°, 72.4°)	(44.7°, 75.6°)	(47.8°, 77.1°)
HAQPSO	(40.8°, 70.9°)	(41.4°, 72.6°)	(44.4°, 73.5°)

TABLE 3. Proximity effect analysis under low SNR for 4 targets.

Alg.	T1(β_1, α_1)	T2(β_2, α_2)	T3(β_3, α_3)	T4(β_4, α_4)
True	40°, 70°	42°, 72°	44°, 74°	46°, 76°
MUSIC	41.8°, 67.3°	44.4°, 74.8°	46.9°, 76.4°	48.7°, 78.2°
TLS-ESPRIT	42.1°, 66.8°	45.0°, 75.4°	47.6°, 77.2°	49.3°, 79.1°
OMP	40.9°, 69.2°	43.5°, 73.1°	45.8°, 75.8°	47.6°, 77.4°
PSO	37.1°, 72.4°	44.7°, 75.6°	47.8°, 77.1°	50.2°, 79.3°
HAQPSO	40.1°, 70.2°	42.2°, 72.1°	44.3°, 74.2°	46.1°, 76.2°

TABLE 4. Estimation error vs snapshot numbers at SNR = -5 dB.

Snapshots	2D-MUSIC	2D-ESPRIT	2D-OMP	HAQPSO
200	1.8°	2.1°	1.2°	0.8°
100	1.9°	2.1°	1.4°	0.8°
50	3.2°	4.1°	4.7°	0.9°
10	5.7°	6.1°	6.7°	0.9°

TABLE 5. Computational complexity and execution time for different algorithms.

Algorithm	Computational Complexity	Execution Time (s)
2D-MUSIC	$\mathcal{O}(M^2L + M^3 + G_\theta G_\phi M(M - K))$	0.13
2D TLS-ESPRIT	$\mathcal{O}(M^2L + M^3 + K^3)$	0.04
2D-OMP	$\mathcal{O}(M^2L + KMG_\theta G_\phi + K^3)$	2.42
PSO	$\mathcal{O}(M^2L + M^3 + T_{\max} N_p K M(M - K))$	0.73
Proposed HAQPSO	$\mathcal{O}(M^2L + M^3 + T_{\max} N_p K M(M - K))$	0.38

where SNR = -5dB. As provided in Table.2, the performance of HAQPSO degraded a bit but still achieved fairly adequate accuracy even in the presence of low SNR, while a notable deviation can be seen in the resolution of the PSO and MUSIC algorithms. MUSIC exhibits significant inaccuracies due to the coarse grid resolution and the sensitivity to noise.

In Table.3, the effect of angular separation is evaluated for four targets located close to each other in the presence of low snr of -5dB. The HAQPSO is compared with 2D-MUSIC [20], 2D TLS-ESPRIT [14] and 2D-OMP [15] methods. 200 snapshots were used in this experiment for 2D-MUSIC, 2D TLS-ESPRIT and 2D-OMP methods. Four targets were considered whose desired elevation and azimuth angles were ($\beta_1 = 40^\circ, \alpha_1 = 70^\circ$), ($\beta_2 = 42^\circ, \alpha_2 = 72^\circ$), ($\beta_3 = 44^\circ, \alpha_3 = 74^\circ$), ($\beta_4 = 46^\circ, \alpha_4 = 76^\circ$). Even in this adverse case of angular separation of less than two degrees between the corresponding elevation and azimuth angles of four different targets, the proposed HAQPSO performed better compared to the other algorithms.

Case e. Performance Under Different Snapshots Number: In this sub-section, the effect of snapshots number on estimation accuracy is evaluated for HAQPSO, 2D-MUSIC, 2D-TLS ESPRIT and 2D-OMP based methods. For this simulation, the same adverse condition of -5dB SNR and 4 closely spaced targets were considered whose true values

were ($\beta_1 = 40^\circ, \alpha_1 = 70^\circ$), ($\beta_2 = 42^\circ, \alpha_2 = 72^\circ$), ($\beta_3 = 44^\circ, \alpha_3 = 74^\circ$), ($\beta_4 = 46^\circ, \alpha_4 = 76^\circ$). As provided in Table.5, the estimation error between the desired angles and estimated angles is significantly increases for MUSIC, ESPRIT and OMP method for less number of snapshots while the HAQPSO maintains a fair less estimation error between true and estimated values.

Case f. Computational Complexity Analysis: The computational complexity of HAQPSO is mainly governed by three computational stages: covariance matrix estimation (performed only once), eigenvalue decomposition (performed only once), and iterative fitness evaluation associated with the swarm optimization process (depending on the number of iterations). The computations required for the sample covariance matrix is $\mathcal{O}(Q^2L)$, while the eigenvalue decomposition (EVD) of the $Q \times Q$ covariance matrix need $\mathcal{O}(Q^3)$ computations. These computations are required only once. Each particle of dimension $2P$ contains the elevation and azimuth angles of all P targets, and therefore at each iteration the objective function is calculated for all M particles. Since the objective evaluation function involves the projection of the noise subspace onto the steering matrix, its main computational cost is approximately $\mathcal{O}(PQ(Q - P))$ and hence the complexity of the optimization stage is $\mathcal{O}(T_{\max} N_p KM(M - K))$. Therefore, the combined theoretical computational cost of HAQPSO is $\mathcal{O}(Q^2L + Q^3 + T_{\max} MPQ(Q - P))$. It is significant to observe that the adaptive quantum update mechanism presents only simple element-wise arithmetic operations over the particle dimension of $2P$. Hence, its computation is negligible compared with repeated fitness evaluation. Therefore, the dominant complexity of HAQPSO is governed by the number of iterations, array size, number of particles and the targets. Table.5 summarizes the real time execution time and theoretical computational complexity of HAQPSO, PSO, MUSIC, OMP and ESPRIT algorithms. It can be observed that theoretically, the computational complexity of the traditional PSO is similar to that of HAQPSO because both algorithms use the same population size. However, since the HAQPSO uses adaptive quantum-behaved position updates and hybridization with a local optimizer, it enhances the exploitation and exploration capabilities of the swarm more effectively compared to traditional PSO. These additional update operations need only simple arithmetic computations and therefore do not change the asymptotic computational complexity. Although both algorithms possess the same Big- $\mathcal{O}$ order, however, HAQPSO require fewer iterations and provide improved convergence characteristics and estimation accuracy compared to PSO. The computational complexity of 2D MUSIC algorithm also consists of three major steps: covariance matrix estimation, EVD and two-dimensional spectrum search. The overall computational complexity of the 2D-MUSIC algorithm is $\mathcal{O}(Q^2L + Q^3 + G_\beta G_\alpha Q(Q - P))$. The prevalent computational burden of 2D-MUSIC arises from the exhaustive two-dimensional angular search and accordingly the computational cost increases substantially as the search-grid resolution becomes finer. Although 2D-MUSIC

provides high-resolution DOA estimation, its complexity increases proportionally with the product $G_\beta G_\alpha$, making it computationally demanding for practical high-resolution 2D DOA estimation problems.

Unlike 2D-MUSIC, the two-dimensional Total Least Squares Estimation of Signal Parameters via Rotational Invariance Techniques (2D TLS-ESPRIT) avoids exhaustive spectral search and directly computes the angular parameters from the signal subspace, significantly minimizes the computational burden. The total computational complexity of the 2D TLS-ESPRIT algorithm is approximated to be $\mathcal{O}(Q^2L + Q^3 + P^3)$. The complexity of the 2D TLS-ESPRIT algorithm is free of search-grid resolution (G_β, G_α) and considered to be one of the computationally efficient DOA estimation techniques. However, its estimation accuracy can be degraded under fewer snapshots, low-SNR condition, and closely spaced targets, due to the sensitivity of the rotational invariance property to subspace estimation errors. The two-dimensional Orthogonal Matching Pursuit (2D-OMP) algorithm considers the DOA estimation problem as a sparse signal reconstruction task. The algorithm uses a predefined dictionary within the search region consisting of AMV corresponding to all possible combinations of elevation and azimuth angles. The Computational burden associated with the dictionary construction stage is $\mathcal{O}(QG_\beta G_\alpha)$ and the total complexity of the 2D-OMP is $\mathcal{O}(Q^2L + Q(G_\beta G_\alpha + P) + P^3)$. In order to get a better estimate through OMP, a large dictionary must be stored and must be searched repeatedly during each iteration. In general, 2D-OMP provides improved resolution compared with MUSIC and ESPRIT, however, its computational cost and memory requirements increase substantially as the search space expands.

In the same way, the real time execution analysis shows that 2D TLS-ESPRIT is the most computationally efficient method, whereas 2D-MUSIC and 2D-OMP become expensive as the search-grid resolution and dictionary size increases respectively. Although the proposed HAQPSO requires additional computational effort due to iterative optimization, it eliminates exhaustive two-dimensional spectral search and provides a favourable trade-off between computational complexity, estimation accuracy, and robustness under adverse conditions such as low-SNR environments and closely spaced targets.

V. CONCLUSION AND FUTURE RECOMMENDATIONS

In this work, a novel methodology based on the hybridization of QPSO and deterministic coordinate descent (DCD) algorithm was developed for the joint 2D DOA estimation of far field targets received on rectangular array. The performance of HAQPSO was established in terms of estimation accuracy for different number of targets and Its performance was compared with PSO and MUSIC, ESPRIT and OMP based algorithms under adverse conditions, such as for closely spaced targets, limited number of sanpshots availability and low SNR conditions. The performance of the proposed hybrid scheme was observed to be

comparable to that of the MUSIC algorithm, ESPRIT and OMP methods especially in adverse environments. The superior performance of HAQPSO is primarily attributed to its multi-source fitness formulation, efficient exploration of the search space, and robust handling of the noise subspace projection. It should be noted that practical array systems may suffer from additional adverse conditions such as mutual coupling among the sensors of array, sensor calibration errors, phase uncertainties and gain [59], [60]. Such adverse effects can distort the array manifold and spoiled the performance of DOA estimator techniques. Since the proposed HAQPSO algorithm operates by optimizing a cost function defined through the array manifold matrix, these adverse conditions can be implemented into the signal model by incorporating an appropriate sensor calibration or error-aware manifold vector formulation. Hence, the proposed framework can be extended in future to jointly estimate the target parameters in the presence of these adverse conditions. Further, one can develop the same algorithm for Bi-static MIMO radar system where the direction of departure is not equal to the direction of arrival. In the same way, one can use the same methodology for generating nulls in the direction of interferences and produce main beam in the desired direction.

APPENDIX

A. PSEUDO-CODE OF HAQPSO FOR 2D DOA ESTIMATION

- 1: **Input:** Number of particles M , maximum iterations T , number of sources P , initial quantum coefficient γ_0 , covariance $\mathbf{R}$, snapshots L .
- 2: **Output:** Estimated DOAs $\hat{\beta}, \hat{\alpha}$.
- 3: Initialize particle population $\mathbf{p}_m$ randomly for $m = 1, \dots, M$.
- 4: Compute noise subspace $\mathbf{U}_n$ from $\mathbf{R}$.
- 5: **for** $t = 1$ to T **do**
- 6: Compute cost $J(\mathbf{p}_m)$ for all particles.
- 7: Update $\mathbf{m}_{\text{best}} = \frac{1}{M} \sum_m \mathbf{p}_m$.
- 8: **for** each particle m **do**
- 9: Update particles using quantum update: $\mathbf{p}_m \leftarrow \mathbf{i}_{\text{best}} + \text{sign}(\mathbf{r}) \cdot \gamma(t) \cdot |\mathbf{i}_{\text{best}} - \mathbf{p}_m| \cdot \ln(1/\mathbf{u})$.
- 10: Reflect particles within angular bounds.
- 11: **end for**.
- 12: Update elite: $\mathbf{p}_{\text{elite}} = \arg \min_m J(\mathbf{p}_m)$.
- 13: **end for**.
- 14: Apply deterministic local refinement on $\mathbf{p}_{\text{elite}}$.
- 15: Return $\hat{\beta}, \hat{\alpha}$ from refined elite particle.

REFERENCES

- [1] S. Kulkarni, A. Thakur, S. Soni, A. Hiwale, M. H. Belsare, and A. A. B. Raj, "A comprehensive review of direction of arrival (DoA) estimation techniques and algorithms," *J. Electron. Electr. Eng.*, vol. 4, no. 1, pp. 138–186, Feb. 2025.
- [2] A. M. A. Abdo, M. H. M. Nerma, and A. M. H. Mohamed, "Adaptive direction of arrival estimation and beamforming algorithms for future networks and advancements," *WSEAS Trans. Electron.*, vol. 16, pp. 33–45, Feb. 2025.

- [3] F. Zaman, S. Akbar, Z. Akhtar, and G. Phnomchoeng, "Efficient technique for joint parameter estimation of mixed far field targets and near field jammers in phased array radar," *IEEE Access*, vol. 13, pp. 55887–55898, 2025.
- [4] N. Ahmed, H. Wang, M. A. Z. Raja, W. Ali, M. Kashif, and F. Zaman, "Novel design of grey wolf optimization heuristics for high resolution direction of arrival estimation in acoustic plane waves," *Wireless Pers. Commun.*, vol. 128, no. 4, pp. 2507–2529, Feb. 2023.
- [5] A. Mehrabian and G. Kaddoum, "Enhancing resilience against jamming attacks: A cooperative anti-jamming method using direction estimation," *IEEE Trans. Commun.*, vol. 73, no. 11, pp. 12696–12710, Nov. 2025.
- [6] J.-S. Row, C.-Y. Lin, and C.-H. Chuang, "Direction of arrival estimation system based on a vertically-polarized circular array with 360° continuous null steering," *IEEE Wireless Commun. Lett.*, vol. 14, no. 9, pp. 2843–2847, Sep. 2025.
- [7] G. Antonelli et al., "Towards a multi-robot architecture for bearing-only tracking of underwater acoustic sources using vector sensors," *Ocean Eng.*, vol. 353, Apr. 2026, Art. no. 124714.
- [8] S. Akbar, F. Zaman, R. Ullah, N. Gul, A. B. Alhassan, and G. Phnomchoeng, "Direction of arrival estimation in the presence of imperfect waveforms for multiple targets in MIMO radar," *IEEE Access*, vol. 12, pp. 164262–164273, 2024.
- [9] R. Schmidt, "Multiple emitter location and signal parameter estimation," *IEEE Trans. Antennas Propag.*, vol. AP-34, no. 3, pp. 276–280, Mar. 1986.
- [10] F. Zaman, Z. Akhtar, S. Akbar, and G. Phnomchoeng, "Joint signal-power and direction of arrival estimation for far-field targets using novel metaheuristic framework," *IEEE Access*, vol. 14, pp. 65923–65931, 2026.
- [11] S. Akbar, M. A. Zahoor Raja, N. I. Chaudhary, F. Zaman, and H. Alquhayz, "Flower pollination heuristics for parameter estimation of electromagnetic plane waves," *Comput., Mater. Continua*, vol. 68, no. 2, pp. 2529–2543, 2021.
- [12] J. An et al., "Two-dimensional direction-of-arrival estimation using stacked intelligent metasurfaces," *IEEE J. Sel. Areas Commun.*, vol. 42, no. 10, pp. 2786–2802, Oct. 2024.
- [13] F. Zaman, A. Iqbal, B. Ali, and A. K. J. Saudagar, "Joint estimation of elevation and azimuth angles with triple-parallel ULAs using metaheuristic and direct search methods," *Comput. Model. Eng. Sci.*, vol. 145, no. 2, pp. 2535–2550, 2025.
- [14] L. Miao, S. Zhang, L. Huang, J. Ding, and W.-Q. Wang, "Moving target detection using FDA-MIMO radar with planar array," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.
- [15] C. Wang, J. Hu, Q. Zhang, and X. Yuan, "An efficient 2D DOA estimation algorithm based on OMP for rectangular array," *Electronics*, vol. 12, no. 7, p. 1634, Mar. 2023.
- [16] S. Chen et al., "Near-field communications: Shape and structure design for uniform planar array," *IEEE Internet Things J.*, vol. 12, no. 11, pp. 15151–15165, Jun. 2025.
- [17] M. Suga, Y. Takahashi, Y. Yamamoto, H. Toshinaga, N. Kita, and T. Onizawa, "2D pre-beamforming control method via analog radio over fiber for wireless communication systems with massive planar array antenna," *IEEE Access*, vol. 13, pp. 61760–61767, 2025.
- [18] C. Yuxin, X. Jian, Z. Qiangqiang, L. Qiying, Y. Dewen, and H. Jun, "An efficient asymmetric hysteresis modeling framework for multiple bolted joints with application to large planar array antennas," *Chin. J. Aeronaut.*, vol. 39, no. 8, Aug. 2026, Art. no. 103732.
- [19] J.-W. Lee and S.-J. Yang, "Efficient estimation of active element patterns for 2-D planar array antennas via directional decomposition," *IEEE Antennas Wireless Propag. Lett.*, vol. 24, pp. 3410–3414, 2025.
- [20] Y. Zhao, X. Fan, J. Liu, Y. Li, L. Yao, and J. Wang, "Efficient 2D-DOA estimation based on triple attention mechanism for L-shaped array," *Sensors*, vol. 25, no. 8, p. 2359, Apr. 2025.
- [21] H. Xu, W. Liu, H. Chen, and Y. Tian, "Efficient 2-D DOA estimation for incoherently distributed sources with an L-shaped array," *IEEE Wireless Commun. Lett.*, vol. 14, no. 4, pp. 1104–1108, Apr. 2025.
- [22] L. Yao, R. Zhang, C. Hu, C. Ye, W. Wu, and B. Shim, "Tensor-based 2-D DOA estimation for uniform planar arrays with unknown mutual coupling," *IEEE Internet Things J.*, vol. 13, no. 3, pp. 3899–3912, Feb. 2026.
- [23] F. H. Shah, M. A. Z. Raja, F. Al-Turjman, F. Zaman, and X. Yang, "Evolutionary heuristic computing paradigm for 2D-DOA estimation along circular array," *Wireless Commun. Mobile Comput.*, vol. 2022, no. 1, Jan. 2022, Art. no. 4851364.
- [24] Y. Yue et al., "2D DOA estimation for multiple parallel sparse linear arrays via recursive roots finding," *IEEE Trans. Veh. Technol.*, vol. 74, no. 11, pp. 18226–18231, Nov. 2025.
- [25] F. Taga, "Smart music algorithm for DOA estimation," *Electron. Lett.*, vol. 33, no. 3, pp. 190–191, 1997.
- [26] F.-M. Han and X.-D. Zhang, "An ESPRIT-like algorithm for coherent DOA estimation," *IEEE Antennas Wireless Propag. Lett.*, vol. 4, pp. 443–446, 2005.
- [27] F. Li, H. Liu, and R. J. Vaccaro, "Performance analysis for DOA estimation algorithms: Unification, simplification, and observations," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 29, no. 4, pp. 1170–1184, Oct. 1993, doi: 10.1109/7.259520.
- [28] H. Abeida and J.-P. Delmas, "MUSIC-like estimation of direction of arrival for noncircular sources," *IEEE Trans. Signal Process.*, vol. 54, no. 7, pp. 2678–2690, Jul. 2006.
- [29] J. Liang and D. Liu, "Passive localization of mixed near-field and far-field sources using two-stage MUSIC algorithm," *IEEE Trans. Signal Process.*, vol. 58, no. 1, pp. 108–120, Jan. 2010.
- [30] B. D. Rao and K. V. S. Hari, "Performance analysis of root-music," *IEEE Trans. Acoust., Speech, Signal Process.*, vol. 37, no. 12, pp. 1939–1949, Dec. 1989, doi: 10.1109/29.45540.
- [31] Z. Yang, "Nonasymptotic performance analysis of ESPRIT and spatial-smoothing ESPRIT," *IEEE Trans. Inf. Theory*, vol. 69, no. 1, pp. 666–681, Jan. 2023.
- [32] C. Li, L. Xie, Z. Liu, B. Zhou, and Q. Ma, "A subspace spatial smoothing-based sparse reconstruction passive direction-of-arrival estimation method under strong interference," *J. Acoust. Soc. Amer.*, vol. 157, no. 4, pp. 2376–2391, Apr. 2025.
- [33] X. Zhang, D. Feng, and W. Zheng, "Fast subspace and DOA estimation method for the case of high-dimensional and small samples," *IEEE Trans. Veh. Technol.*, vol. 74, no. 3, pp. 3958–3975, Mar. 2025.
- [34] T. A. El-Rahman, A. A. Salama, W. M. Saad, and M. H. El-Shafey, "Cramér–Rao bound investigation of a novel virtually extended nested arc array for 2D DOA estimation," *Alexandria Eng. J.*, vol. 76, pp. 259–274, Aug. 2023.
- [35] X. Li, W. Wang, X. Wang, and S. Ren, "Synthetic sparse planar array design for two-dimensional DOA estimation," *Digit. Signal Process.*, vol. 120, Jan. 2022, Art. no. 103268.
- [36] M. Li and Y. Lu, "A refined genetic algorithm for accurate and reliable DOA estimation with a sensor array," *Wireless Pers. Commun.*, vol. 43, no. 2, pp. 533–547, Oct. 2007.
- [37] P. Geetha, S. J. Nanda, and R. P. Yadav, "A parallel chaotic sailfish optimization algorithm for estimation of DOA in wireless sensor array," *Phys. Commun.*, vol. 51, Apr. 2022, Art. no. 101536.
- [38] M. Fan, J. Zhang, M. Li, and L. Qiao, "Gridless cooperative coevolutionary algorithm for real-time DOA estimation in automotive radar," *IEEE Trans. Instrum. Meas.*, vol. 75, pp. 1–16, 2026.
- [39] C. Lan, H. Chen, L. Zhang, R. Guo, C. Han, and D. Luo, "Underwater acoustic DOA estimation of incoherent signal based on improved GA-MUSIC," *IEEE Access*, vol. 11, pp. 69474–69485, 2023.
- [40] F. Zaman, "Joint angle-amplitude estimation for multiple signals with L-structured arrays using bioinspired computing," *Wireless Commun. Mobile Comput.*, vol. 2017, pp. 1–12, Jan. 2017. [Online]. Available: <https://doi.org/10.1155/2017/9428196>
- [41] F. Zaman, I. M. Qureshi, F. Munir, and Z. U. Khan, "Four-dimensional parameter estimation of plane waves using swarming intelligence," *Chin. Phys. B*, vol. 23, no. 7, Jul. 2014, Art. no. 078402.
- [42] T.-P. Chang and C.-L. Meng, "Efficient direction of arrival estimation using particle swarm optimization for hybrid analog and digital massive multiple-input multiple-output receiving array," *Sensors Mater.*, vol. 37, no. 3, p. 861, 2025.
- [43] S. Hakemi, M. Houshmand, E. KheirKhah, and S. A. Hosseini, "A review of recent advances in quantum-inspired metaheuristics," *Evol. Intell.*, vol. 17, no. 2, pp. 627–642, Apr. 2024.
- [44] Pooja and S. K. Sood, "Scientometric analysis of quantum-inspired metaheuristic algorithms," *Artif. Intell. Rev.*, vol. 57, no. 2, p. 22, Jan. 2024.
- [45] A. Gunjan and S. Bhattacharyya, "Quantum-inspired meta-heuristic approaches for a constrained portfolio optimization problem," *Evol. Intell.*, vol. 17, no. 4, pp. 3061–3100, Aug. 2024.
- [46] M. O. Butt, N. Waheed, T. Q. Duong, and W. Ejaz, "Quantum-inspired resource optimization for 6G networks: A survey," *IEEE Commun. Surveys Tuts.*, vol. 27, no. 5, pp. 2973–3019, Oct. 2025.
- [47] S. Fallahi and M. Taghadosi, "Quantum-behaved particle swarm optimization based on solitons," *Sci. Rep.*, vol. 12, no. 1, p. 13977, Aug. 2022.
- [48] E. Baş, "Improved particle swarm optimization on based quantum behaved framework for big data optimization," *Neural Process. Lett.*, vol. 55, no. 3, pp. 2551–2586, Jun. 2023.

- [49] W. Velasquez, F. Jijon-Veliz, and M. S. Alvarez-Alvarado, "Optimal wireless sensor networks allocation for wooded areas using quantum-behaved swarm optimization algorithms," *IEEE Access*, vol. 11, pp. 14375–14384, 2023.
- [50] S. J. Wright, "Coordinate descent algorithms," *Math. Program.*, vol. 151, no. 1, pp. 3–34, Jun. 2015.
- [51] Y. Yang, L. Xue, and G. Li, "Adaptive elastic net penalized high-dimensional quantile regression models with generalized coordinate descent algorithm," *Stat. Papers*, vol. 67, no. 1, p. 9, Feb. 2026.
- [52] K.-H. Kim et al., "Development of planar active phased array antenna for detecting and tracking radar," in *Proc. IEEE Radar Conf. (RadarConf18)*, Apr. 2018, pp. 100–103.
- [53] T. C. Cheston and J. Frank, *Phased Array Radar Antenna*. New York, NY, USA: McGraw-Hill, 1990.
- [54] J. Kennedy and R. Eberhart, "Particle swarm optimization (PSO)," in *Proc. IEEE Int. Conf. Neural Netw.*, Dec. 1995, pp. 1942–1948.
- [55] B. Nakisa, M. Z. A. Nazri, M. N. Rastgoo, and S. Abdullah, "A survey: Particle swarm optimization based algorithms to solve premature convergence problem," *J. Comput. Sci.*, vol. 10, no. 9, pp. 1758–1765, 2014.
- [56] M. Haris and H. Nam, "Path planning optimization of smart vehicle with fast converging distance-dependent PSO algorithm," *IEEE Open J. Intell. Transp. Syst.*, vol. 5, pp. 726–739, 2024.
- [57] M. Xi, J. Sun, and W. Xu, "An improved quantum-behaved particle swarm optimization algorithm with weighted mean best position," *Appl. Math. Comput.*, vol. 205, no. 2, pp. 751–759, Nov. 2008.
- [58] S. Dohare and R. S. Rajput, "Adaptive Gaussian quantum based PSO and TSA optimization for parametric optimization of toughened glass on toughening machine," *Ceram. Int.*, vol. 48, no. 16, pp. 22799–22807, Aug. 2022.
- [59] Y. Yue, Y. Wang, F. Xing, Z. Shi, and G. Liao, "Polynomial rooting-based parameter estimation for polarimetric monostatic MIMO radar," *Signal Process.*, vol. 212, Nov. 2023, Art. no. 109172.
- [60] Y. Yue, Y. Xu, Z. Liu, and L. Shen, "Parameter estimation of coexisted circular and strictly noncircular sources using diversely polarized antennas," *IEEE Commun. Lett.*, vol. 22, no. 9, pp. 1822–1825, Sep. 2018.



FAWAD ZAMAN (Senior Member, IEEE) received the M.S. and Ph.D. degrees in electronic engineering from International Islamic University Islamabad, Pakistan, in 2009 and 2013, respectively. Since September 2025, he has been a Post-Doctoral Researcher with the Department of Mechanical Engineering, Chulalongkorn University, Bangkok, Thailand, under the C2F Post-Doctoral Fellowship Program. From 2021 to 2022, he was a Visiting Researcher with the Department of Electrical and Electronic Engineering, Imperial College London,

U.K. His research interests include array signal processing, wireless communications, beamforming, direction-of-arrival (DOA) estimation, antenna arrays, MIMO radar systems, and optimization techniques. He has served as a lead guest editor for several notable special issues.



SADIQ AKBAR received the Ph.D. degree in electronics from the University of Peshawar, Khyber Pakhtunkhwa, Pakistan, in 2019. He has been an integral part of the Department of Electronics, University of Peshawar, as a Lecturer, since 2000, and advancing to the role of an Assistant Professor in 2010. He completed his post-doctoral research at the Department of Mechanical Engineering, Chulalongkorn University, Bangkok, Thailand, in 2023, where he contributed to advanced research and innovation in his field. He is a prolific author,

having contributed to several research articles published in reputable journals. His expertise and dedication have also led him to review several international journal publications and serve as the Session Chair for International Conference on Electrical, Communication, and Computer Engineering (ICECCE2019). His commitment to international collaboration has significantly enhanced the quality and impact of his research, fostering the integration of diverse perspectives and resources. His main research interests encompass a broad range of topics, including array signal processing, parameter estimation, direction of arrival estimation, antenna arrays, MIMO radars, system identification, evolutionary computing, swarm intelligence, nonlinear PID controller design, beamforming, and optimization methods.



AMIR RASHID CH received the B.A. degree in journalism from Punjab University in 1995, the master's degree in political science from Bahawalpur Islamia University in 1998, the Bachelor of Science degree in electrical engineering and the Master of Science degree in computer sciences from the University of Engineering and Technology (UET), Lahore, in 2000 and 2003, respectively, and the Ph.D. degree from Lancaster University, U.K., in 2014, under the esteemed Lancaster Chancellor Scholarship. He stands as a beacon of excellence in academia, research, and innovation, celebrated worldwide for his remarkable contributions to electrical engineering, computer sciences, and communication systems. With over 22 years of combined industrial and teaching experience, he is currently an Associate Professor with COMSATS University, Islamabad, Pakistan. His groundbreaking research has earned him national and international acclaim, with prestigious awards from the President of Pakistan and the Federation of Pakistan Chambers of Commerce and Industry (FPCCI). His diverse research portfolio spans network engineering, medical research, communication systems, control systems, and artificial intelligence, catalyzing significant advancements in these fields. A visionary leader in STEM education and research, he inspires and mentors students and researchers, leaving an indelible mark on the landscape of science and technology.



SHAFQAT ULLAH KHAN received the M.S. degree in electronic engineering from International Islamic University Islamabad in 2008 and the Ph.D. degree in electronic engineering from ISRA University in 2015. He was a Post-Doctoral Fellow with the Faculty of Electrical Engineering, University Technology Malaysia, from 2016 to 2017. He is an Associate Professor with the University of Buner. His research interests include the RF and microwave, antenna arrays, and evolutionary algorithms.



GRIDSADA PHANOMCHOENG (Senior Member, IEEE) received the B.S. degree in mechanical Engineering from Chulalongkorn University, Bangkok, Thailand, in 2002, and the M.S. degree in aerospace engineering and mechanics and the Ph.D. degree in control science and dynamical systems from the University of Minnesota, Twin Cities, Minneapolis, MN, USA, in 2007 and 2011, respectively. In 2012, he was a Post-Doctoral Researcher with the Dr. Rajamani's Mechanical Engineering Laboratory, University of Minnesota, Twin Cities. Currently, he is a Faculty Member of mechanical engineering with Chulalongkorn University. His research interests include advanced control system design, observer design for nonlinear systems, system identification, applications to automotive systems, energy harvesting, and machine vision.